

Requirements Report

AI Interview Preparation Platform

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1. Project Overview

The **AI Interview Preparation Platform** is a full-stack, AI-powered web application designed to help candidates systematically prepare for technical and HR interviews. It leverages HuggingFace language models to simulate real interview environments, provide personalized feedback, analyze resumes, and track user progress.

2. Stakeholders

Stakeholder	Role	Interest
Job Seekers / Students	Primary Users	Prepare for technical and HR interviews
Recruiters / HR Teams	Secondary Users	Evaluate platform-prepared candidates
Platform Administrators	Admin	Manage content, users, and AI models
Development Team	Builders	Design, build, and maintain the platform

3. Business Requirements

ID	Requirement
BR-01	The platform must provide AI-driven, personalized interview preparation

BR-02	The system must support multiple interview categories (DSA, System Design, HR, Core CS)
BR-03	The platform must be accessible via modern web browsers without installation
BR-04	Users must be able to track their progress over time
BR-05	The system must integrate with HuggingFace AI models for question generation and evaluation

4. Functional Requirements

4.1 User Authentication

ID	Requirement	Priority
FR-01	Users shall register with email and password	High
FR-02	Users shall log in and receive a JWT token for session management	High
FR-03	Users shall update their profile information	Medium
FR-04	Passwords shall be hashed using bcryptjs before storage	High

4.2 AI-Generated Interview Questions

ID	Requirement	Priority
FR-05	The system shall generate questions based on topic, difficulty, and job role	High
FR-06	Questions shall cover DSA, System Design, Behavioral, and HR categories	High
FR-07	The AI model shall evaluate user answers and provide feedback	High
FR-08	Questions shall adapt to the candidate's skill level and past performance	Medium

4.3 Interview Simulation

ID	Requirement	Priority
FR-09	The system shall simulate a real interview with timed questions	High
FR-10	Users shall submit answers via text input	High
FR-11	Users shall be able to submit answers via voice recording	Medium
FR-12	The system shall provide a post-session AI debrief with scoring	High

4.4 Resume Analysis

ID	Requirement	Priority
FR-13	Users shall upload resumes in PDF format	High
FR-14	The AI shall parse and analyze the resume content	High
FR-15	The system shall identify strengths, gaps, and ATS keyword mismatches	High
FR-16	Users shall receive actionable improvement suggestions	Medium

4.5 Question Bank

ID	Requirement	Priority
FR-17	The platform shall maintain a curated, filterable question bank	High
FR-18	Questions shall be filterable by topic, difficulty, and category	High
FR-19	Users shall practice individual questions outside of interview mode	Medium

4.6 Progress Tracking and Analytics

ID	Requirement	Priority
FR-20	The system shall display a dashboard with progress metrics	High
FR-21	Analytics shall include performance per topic, time spent, and accuracy	High
FR-22	The system shall identify and highlight weak areas	Medium

FR-23	Users shall see study streak information	Low
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4.7 Mock Interview Scheduling

ID	Requirement	Priority
FR-24	Users shall schedule full-length mock interviews for a future time	Medium
FR-25	The system shall notify users before a scheduled session	Low

5. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	API responses shall complete within 2 seconds for standard requests
NFR-02	Performance	AI model responses shall complete within 10 seconds
NFR-03	Scalability	The backend shall support horizontal scaling via stateless JWT auth
NFR-04	Security	All API endpoints (except auth) shall require JWT verification
NFR-05	Security	User passwords shall never be stored in plaintext
NFR-06	Security	File uploads shall be restricted to PDF, max 5MB
NFR-07	Availability	The platform shall target 99.5% uptime
NFR-08	Usability	The UI shall be fully responsive across desktop and mobile
NFR-09	Maintainability	Code shall follow SRP with Routes → Controllers → Services → Models layering

NFR-10	Compatibility	The frontend shall support the last 2 major versions of Chrome, Firefox, and Edge
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6. System Constraints

- The AI functionality is dependent on HuggingFace Inference API availability.
- MongoDB is required as the database; relational databases are not supported.
- Voice-based answer input requires browser microphone permissions.
- Resume analysis is limited to PDF file format.

7. Assumptions

- Users have a stable internet connection.
- Users have a valid HuggingFace API key configured in the environment.
- The platform is deployed in a Node.js 18+ runtime environment.
- MongoDB instance (local or Atlas) is available and accessible.

8. Dependencies

Dependency	Version	Purpose
React	18.x	Frontend SPA framework
Node.js	20.x	Backend runtime
Express.js	4.x	RESTful API server
MongoDB / Mongoose	7.x	Database and ODM
HuggingFace Inference API	Latest	AI question generation and evaluation
JWT / bcryptjs	Latest	Auth and security
Multer / PDF-Parse	Latest	Resume upload and parsing

End of Requirements Report